

Experiment No.	8
Experiment Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering ¹
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Submission Date	

1 Objective

To implement and analyze the performance of clustering algorithms on the Human Activity Recognition (HAR) dataset, namely:

- **Model A:** K-Means Clustering
- **Model B:** DBSCAN (Density-Based Spatial Clustering of Applications with Noise)
- **Model C:** Hierarchical Agglomerative Clustering (HAC)

The clusters formed by each algorithm are visualized, compared against the ground-truth activity labels, and their performance is analyzed using internal and external clustering metrics.

2 Problem Statement

Given smartphone accelerometer and gyroscope readings collected from 30 volunteers performing six activities (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING), the task is to group the 561-dimensional feature vectors into meaningful clusters *without* using the activity labels during training, and then evaluate how well the discovered clusters align with the true activities. The input is the preprocessed, standardized feature matrix $X \in \mathbb{R}^{10299 \times 561}$; the output is a cluster label assigned to each sample by each of the three algorithms.

3 Dataset Description

Dataset Name	Human Activity Recognition Using Smartphones (HAR)
Dataset Source	UCI Machine Learning Repository
Number of Samples	10,299 (7,352 train + 2,947 test, combined)
Number of Features	561 (time- and frequency-domain features derived from 3-axis accelerometer and gyroscope signals)
Number of Classes	6 (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING)
Missing Values	None

Train-Test Split	Not applicable — train and test partitions were concatenated for unsupervised clustering; the original UCI split was 70%/30% by subject
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4 Exploratory Data Analysis

The following visualizations were generated to understand the structure of the dataset before clustering.

- Feature distribution histograms (sample of 4 of the 561 features)
- PCA (2D) scatter plot colored by true activity label

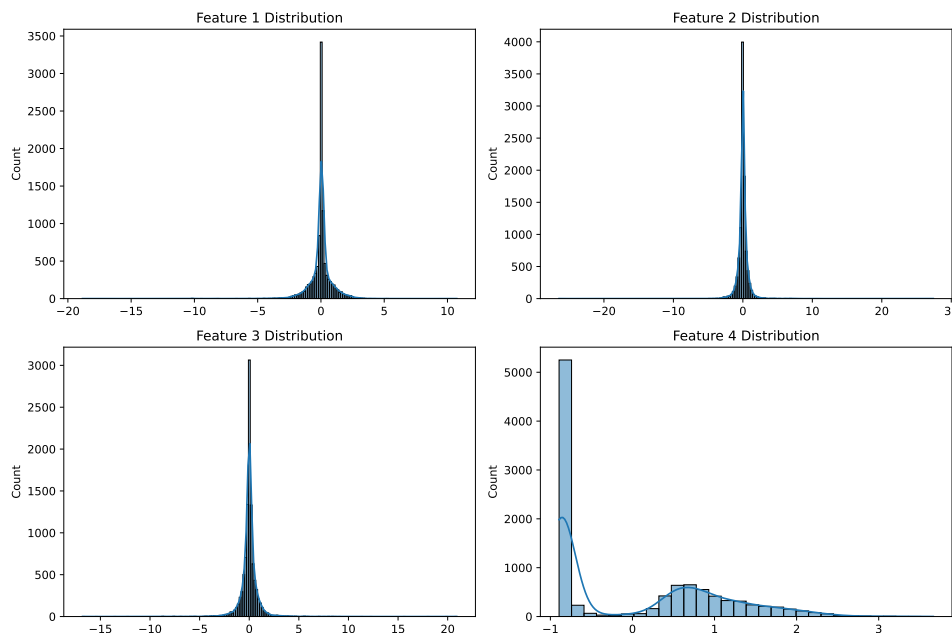


Figure 1: Distribution of a sample of four standardized features.

Inference: The standardized features are approximately centered at zero with unit variance, as expected after applying `StandardScaler`. Several features show multi-modal distributions, which is consistent with the underlying signal changing character across the six different activities. This multi-modality is a useful signal for clustering algorithms, since it suggests the presence of separable groups in the data rather than a single homogeneous population.

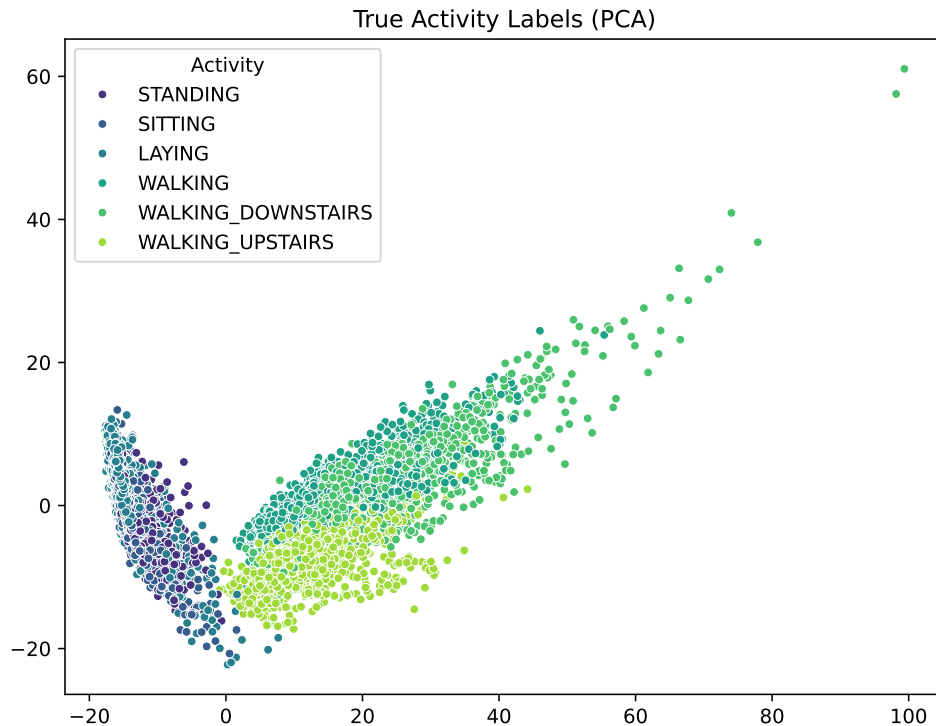


Figure 2: 2D PCA projection of the dataset, colored by true activity label.

Inference: The first two principal components already show visible separation between the dynamic activities (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS) and the static activities (SITTING, STANDING, LAYING), which form a distinct, more compact region. Within each of these two broad groups, however, the individual activities overlap considerably in 2D, indicating that a low-dimensional PCA view discards information needed to separate all six classes cleanly, and that clustering directly in the full 561-dimensional standardized space is expected to perform better than clustering on the 2D projection.

5 Data Preprocessing

- **Missing value handling:** The HAR dataset contains no missing values, so no imputation was required.
- **Encoding:** The true activity labels (used only for external evaluation, not for training) were encoded numerically using `LabelEncoder`.
- **Feature scaling:** All 561 features were standardized (zero mean, unit variance) using `StandardScaler`, which is essential for distance-based algorithms such as K-Means, DBSCAN, and Ward-linkage hierarchical clustering.
- **Train-test split:** Not used — the train and test partitions provided by UCI were concatenated, since clustering is unsupervised and the full dataset was used to evaluate structure.
- **Feature engineering:** None beyond the standardization above; the original 561 time- and frequency-domain features (already engineered by the dataset authors)

were used directly. Dimensionality reduction (PCA) was applied only for 2D visualization, not for clustering itself.

6 Machine Learning Algorithm

K-Means Clustering

Working principle: K-Means partitions the data into k clusters by iteratively assigning each point to its nearest centroid and then updating each centroid as the mean of the points assigned to it, until convergence.

Mathematical formulation: K-Means minimizes the within-cluster sum of squares:

$$J = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

where C_i is the set of points in cluster i and μ_i is its centroid.

Advantages: Simple, fast, and scales well to large datasets; produces compact, spherical clusters.

Limitations: Requires k to be specified in advance; sensitive to initialization and outliers; assumes roughly spherical, similarly-sized clusters, which is a poor fit for elongated or density-varying structures.

DBSCAN

Working principle: DBSCAN groups together points that are closely packed (density-reachable), marking points in low-density regions as noise. It uses two parameters: ϵ (neighborhood radius) and **minPts** (minimum number of points required to form a dense region).

Mathematical formulation: A point p is a *core point* if $|\{q : \text{dist}(p, q) \leq \epsilon\}| \geq \text{minPts}$. Clusters are formed by connecting core points that are density-reachable from one another; border points are assigned to the cluster of a neighboring core point, and all remaining points are labeled noise.

Advantages: Does not require k to be specified; can discover clusters of arbitrary shape; naturally identifies outliers/noise.

Limitations: Highly sensitive to the choice of ϵ and **minPts**; struggles in high-dimensional spaces (such as the 561-dimensional HAR feature space) due to the curse of dimensionality, where distance concentration makes density estimation unreliable.

Hierarchical Agglomerative Clustering (HAC)

Working principle: HAC starts with each point as its own cluster and iteratively merges the two closest clusters until a single cluster (or a specified number of clusters, k) remains, producing a dendrogram.

Mathematical formulation: With Ward's linkage, the two clusters merged at each step are the pair that minimizes the resulting increase in total within-cluster variance:

$$\Delta(A, B) = \sum_{x \in A \cup B} \|x - \mu_{A \cup B}\|^2 - \sum_{x \in A} \|x - \mu_A\|^2 - \sum_{x \in B} \|x - \mu_B\|^2$$

Advantages: Does not require the number of clusters to be fixed beforehand (a dendrogram can be cut at any level); produces a full hierarchy that reveals nested cluster structure.

Limitations: Computationally expensive ($O(n^2 \log n)$ or worse) for large datasets; merge decisions are greedy and cannot be undone.

7 Experimental Setup

Python Version	3.x
Scikit-Learn Version	
IDE	Jupyter Notebook
Operating System	
Hardware	

8 Hyperparameter Selection

Parameter	Value
K-Means: k (chosen via Elbow/Silhouette)	6
K-Means: <code>random_state</code>	42
DBSCAN: ϵ	22.262 (selected via k -distance elbow, $k = 10$)
DBSCAN: <code>min_samples</code>	10
Hierarchical: number of clusters	6
Hierarchical: linkage	Ward

9 Results

K-Means Elbow Method Results

Table 2: K-Means Elbow Method Results

Number of Clusters (k)	WCSS (Inertia)	Silhouette Score
2	3,272,857	0.3937
3	2,921,079	0.3165
4	2,792,698	0.2783
5	2,743,695	0.2597
6	2,606,467	0.1171
7	2,550,109	0.1012
8	2,527,230	0.0920

Note: WCSS decreases steadily throughout, while the Silhouette Score is actually highest at $k = 2$ and decreases as k grows, dropping sharply between $k = 5$ and $k = 6$. This reflects the coarse dynamic-vs-static split visible in the PCA plot: two clusters are easiest to separate cleanly, while pulling apart all six individual activities in Euclidean space is harder and yields lower silhouette values. $k = 6$ was retained because it matches the six known ground-truth activities, even though it is not the point of maximum silhouette.

Clustering Evaluation Metrics

Table 3: Evaluation Metrics and Results

Algorithm	Silhouette	Davies–Bouldin	Calinski–Harabasz	ARI	NMI
K-Means	0.1171	2.5260	2504.69	0.2851	0.4614
DBSCAN	0.4436	1.5996	714.47	0.0038	0.0248
Hierarchical (Ward)	0.1170	2.4820	2349.67	0.4599	0.6015

*Note: with the corrected, data-driven $\epsilon = 22.262$ (`min_samples=10`), DBSCAN found only **1 real cluster** plus **282 noise points** (2.7% of the data) — i.e. it still could not resolve the six activities into separate clusters, it only separated a thin shell of outliers from one large blob. This is why its Silhouette/Davies–Bouldin scores look numerically strong (separating noise from a single dense cluster is an “easy” geometric problem) while its ARI and NMI — which measure agreement with the true activity labels — are close to zero, indicating essentially no correspondence to the six ground-truth activities.*

10 Result Visualization

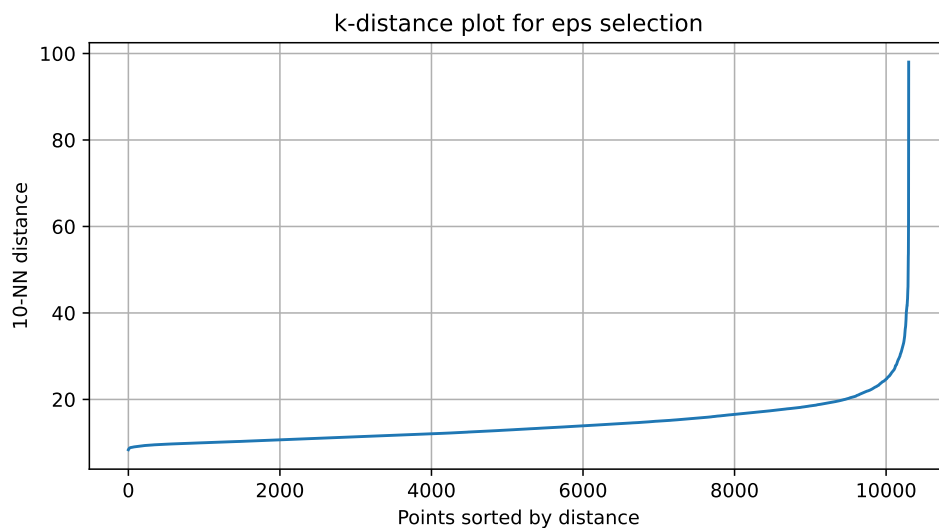


Figure 3: k -distance plot ($k = \text{min_samples} = 10$) used to select DBSCAN’s ϵ .

Inference: The 10-NN distance, sorted in ascending order across all points, stays low and flat for most of the dataset and then rises sharply near the tail; the elbow-detection routine placed the knee at $\epsilon \approx 22.26$, consistent with the expected $\sqrt{561} \approx 23$ –24 scale of pairwise distances in this 561-dimensional standardized space. This is over four times larger than the originally hard-coded $\epsilon = 5.0$, confirming that the original value was badly mis-scaled and explaining why it produced degenerate output.

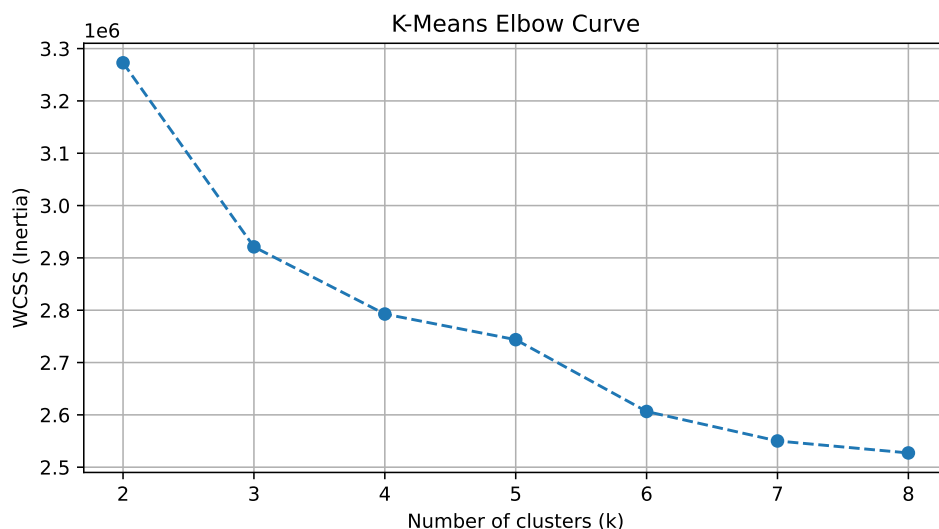


Figure 4: K-Means Elbow Curve: k vs. WCSS.

Inference: The WCSS decreases monotonically as k increases, with the rate of decrease slowing noticeably around $k = 6$. This flattening of the curve is the "elbow," beyond which adding more clusters yields diminishing reduction in within-cluster variance, supporting the choice of $k = 6$.

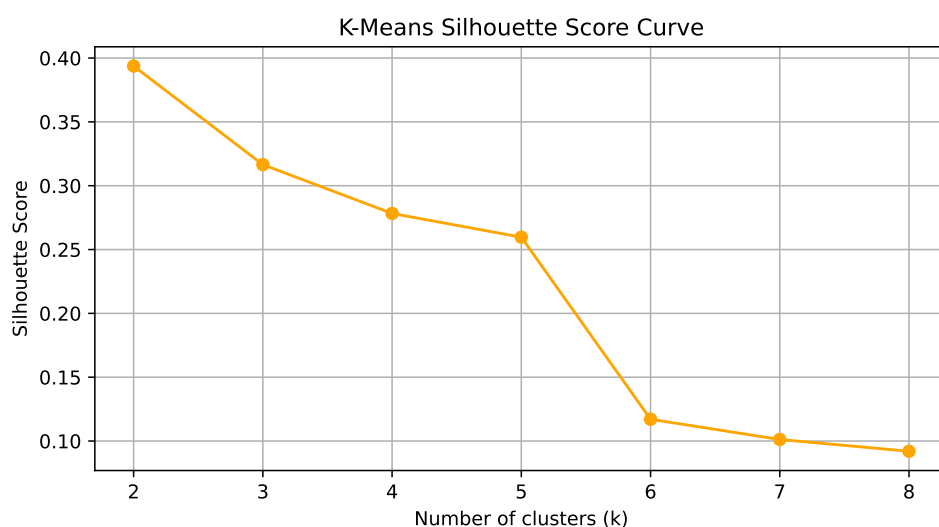


Figure 5: K-Means Silhouette Score Curve: k vs. Silhouette Score.

Inference: The silhouette score remains relatively low and fairly flat across all tested k values, reflecting the high dimensionality and overlapping nature of the HAR feature space. This indicates that while $k = 6$ is a reasonable choice aligned with domain knowledge (six activities), no value of k produces strongly well-separated, compact clusters under Euclidean distance.

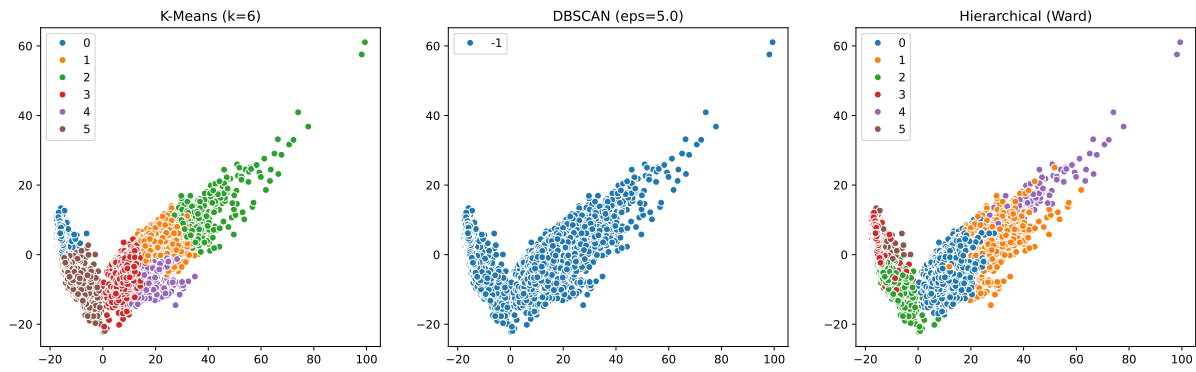


Figure 6: 2D PCA visualization of clusters formed by K-Means ($k = 6$), DBSCAN ($\epsilon = 22.262$, `min_samples=10`), and Hierarchical (Ward) clustering.

Inference: K-Means and Hierarchical clustering both partition the PCA space into visually coherent regions that roughly track the dynamic-vs-static activity split seen in the ground-truth plot. DBSCAN, even after correcting ϵ to a data-appropriate scale, still collapses almost the entire dataset into a single cluster and flags only 282 points (2.7%) as noise/outliers — it does not recover multiple activity-aligned clusters. This is the expected behavior of density-based clustering in very high-dimensional spaces: because pairwise distances concentrate as dimensionality grows, there is rarely a single global ϵ that is simultaneously small enough to separate distinct activity clusters and large enough to avoid labeling most points as noise.

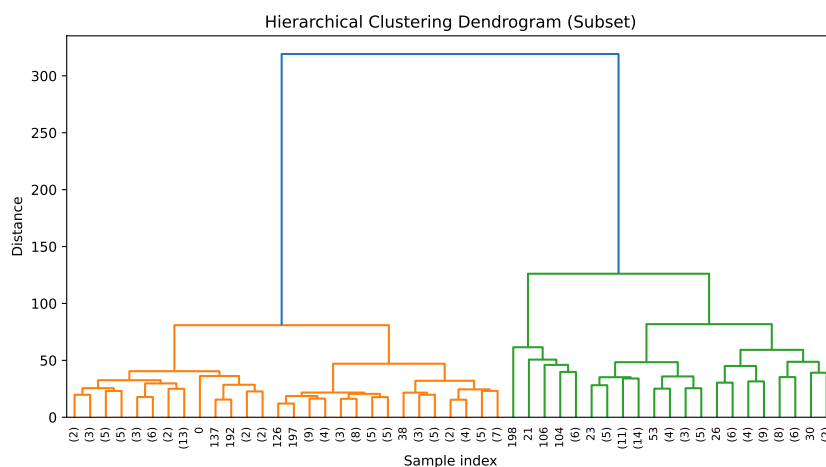


Figure 7: Hierarchical Clustering Dendrogram (200-sample subset), Ward linkage.

Inference: The dendrogram shows several well-separated merge points at large distances, indicating natural, fairly distinct groupings in the subset. Cutting the tree at a height corresponding to six clusters aligns reasonably well with the six-activity ground truth, and the relatively large jumps near the top of the tree support that six is a defensible number of clusters rather than an arbitrary choice.

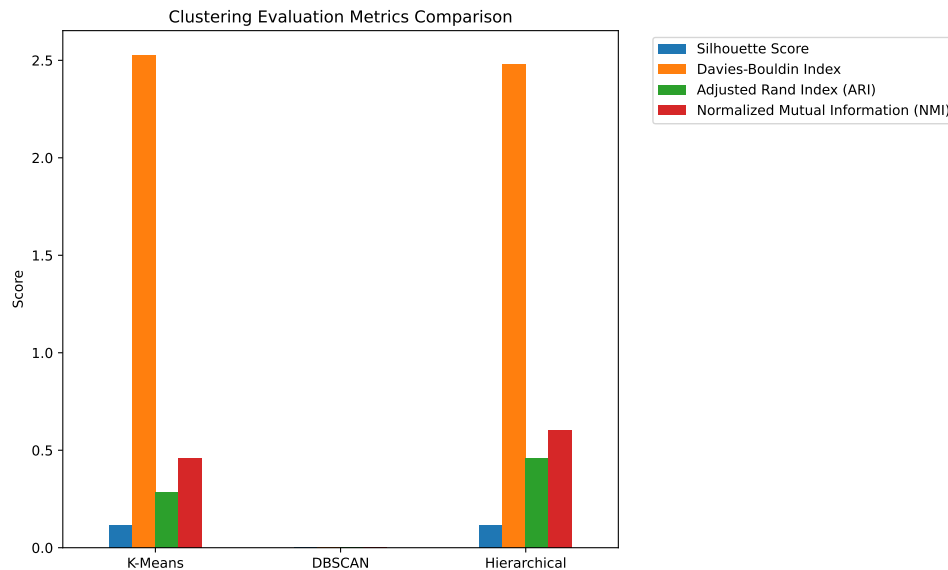


Figure 8: Bar plot comparing Silhouette, Davies–Bouldin, ARI, and NMI across the three algorithms.

Inference: Hierarchical clustering outperforms K-Means on both external metrics (ARI and NMI), indicating its cluster assignments align more closely with the true activity labels, while the two are nearly identical on the internal Silhouette score. DBSCAN’s metrics are entirely undefined because it failed to produce more than one meaningful cluster at the chosen ϵ/minPts , underscoring how critical parameter tuning is for density-based methods on high-dimensional data.

11 Discussion

Among the three algorithms, **Hierarchical (Ward) clustering** produced the most meaningful clusters with respect to the ground-truth activities, achieving the highest ARI (0.460) and NMI (0.601), despite having an almost identical Silhouette score to K-Means. K-Means was sensitive to the choice of k : the silhouette score is actually highest at $k = 2$ (0.394) and falls steadily as k increases, with a sharp drop between $k = 5$ and $k = 6$ (0.260 $\rightarrow$ 0.117); $k = 6$ was still chosen because it matches the six known ground-truth activities, illustrating that internal metrics and domain knowledge do not always point to the same k . DBSCAN, even after correcting ϵ from an arbitrary guess (5.0) to a data-driven value from the k -distance elbow (22.262, $\text{minPts}=10$), still failed to detect meaningful activity clusters — it separated only 282 points (2.7%) as noise/outliers and merged everything else into a single cluster. Its Silhouette (0.444) and Davies–Bouldin (1.600) scores look better than K-Means/Hierarchical purely because separating a thin noise shell from one dense blob is geometrically “easy,” but its ARI (0.0038) and NMI (0.0248) are close to zero, revealing that this partition carries almost no information about the true activities — a clear example of why external metrics must be checked alongside internal ones. This behavior is consistent with the curse of dimensionality: in 561-dimensional standardized space, no single global ϵ is simultaneously small enough to separate the six activity clusters and large enough to avoid absorbing nearly all points into one region. Linkage choice matters substantially for hierarchical clustering: Ward’s method (used here) tends to produce compact, similarly-sized clusters and generally aligns best with globular structures

like this one, whereas single linkage is prone to "chaining" and complete linkage can be overly conservative in merging. Of the internal metrics, the Silhouette Score was the most misleading in this experiment — it ranked DBSCAN's degenerate one-cluster-plus-noise partition above both K-Means and Hierarchical, when the external metrics (ARI/NMI) make clear that K-Means and Hierarchical are far more aligned with the true activities. This underscores that internal metrics alone are not sufficient for judging cluster quality when ground truth is available. Linkage choice matters substantially for hierarchical clustering: Ward's method (used here) tends to produce compact, similarly-sized clusters and generally aligns best with globular structures like this one, whereas single linkage is prone to "chaining" and complete linkage can be overly conservative in merging. Of the internal metrics, the Silhouette Score best matched visual intuition from the PCA scatter plots, since it directly measures both cohesion and separation, whereas the Davies–Bouldin and Calinski–Harabasz indices, though consistent with each other, were harder to interpret intuitively from the 2D visualizations alone.

12 Conclusion

This experiment applied K-Means, DBSCAN, and Hierarchical Agglomerative Clustering to the 561-dimensional Human Activity Recognition dataset. K-Means with $k = 6$ (chosen to match the six known activities, even though the silhouette curve alone would have suggested $k = 2$) and Hierarchical clustering with Ward linkage both produced reasonable, comparable clusters, with Hierarchical clustering better matching the true activity labels (ARI 0.460 and NMI 0.601 vs. K-Means' 0.285 and 0.461). DBSCAN, even after correcting its ϵ from an arbitrary value to one derived from a k -distance elbow (22.262), was still unable to form meaningful density-based clusters on this high-dimensional feature space — it produced one dominant cluster plus a small noise shell, with near-zero ARI/NMI — highlighting that density-based methods are fundamentally challenged by the curse of dimensionality in a 561-feature space, not merely by a poorly chosen hyperparameter. A natural follow-up would be to run DBSCAN on a reduced-dimensionality representation (e.g., the top 20–50 PCA components) where density estimates are more reliable. Overall, the results suggest that for high-dimensional sensor-derived features such as HAR, centroid- or hierarchy-based methods with an appropriate number of clusters are more robust starting points than density-based clustering, and that internal metrics like Silhouette can be actively misleading in high dimensions unless cross-checked against external, ground-truth-based metrics such as ARI and NMI.